signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

*signal.sci

```
// Test Case: Step Response of a Discrete-Time System
// Define the length of the signal
N = 20; // Length of the signal
// Define the input signal (step function)
u_n = ones(1, N); // Step function
// Define the system parameters
b = [1]; // Numerator coefficients of the system (input)
a = [1, -0.5]; // Denominator coefficients of the system (output)
// Compute the system's response to the step input
y_n = filter(b, a, u_n); // System output for step input
// Define the time index for the signals
n = 0:N-1; // Time index
// Plot the step input and system response
clf; // Clear the current figure

// Plot step input
subplot(2,1,1);
bar(n, u_n, 'b');
xlabel('n');
ylabel('u[n]');
title('Step Input Signal');
// Plot system response to step input
subplot(2,1,2);
bar(n, y_n, 'r');
xlabel('n');
ylabel('y[n]');
title('Step Response of the System');
```

